

Cloud Services

Extra Class: MLOps



Tuan Quang – TA

Outline

- Introduction
- Command line (CLI) with Cloud
- Exercise
- Cloud Deploy
- Exercise
- Cloud Monitoring
- Exercise

Introduction

Introduction

❖ What is Cloud

What is Cloud

- Term used to describe a global network of servers, each with a unique function
- The cloud is not a physical entity, but instead is a vast network of remote servers around the globe
- These servers are designed to either store and manage data, run applications, or deliver content or a service such as streaming videos, web mail, office productivity software, or social media



Introduction

❖ Cloud service providers

Topic 6 cloud service providers:

- [Amazon Web Services \(AWS\)](#)
- [Microsoft Azure](#)
- [Google Cloud \(GCP\)](#)
- [Alibaba Cloud](#)
- [IBM Cloud](#)
- [Oracle Cloud](#)

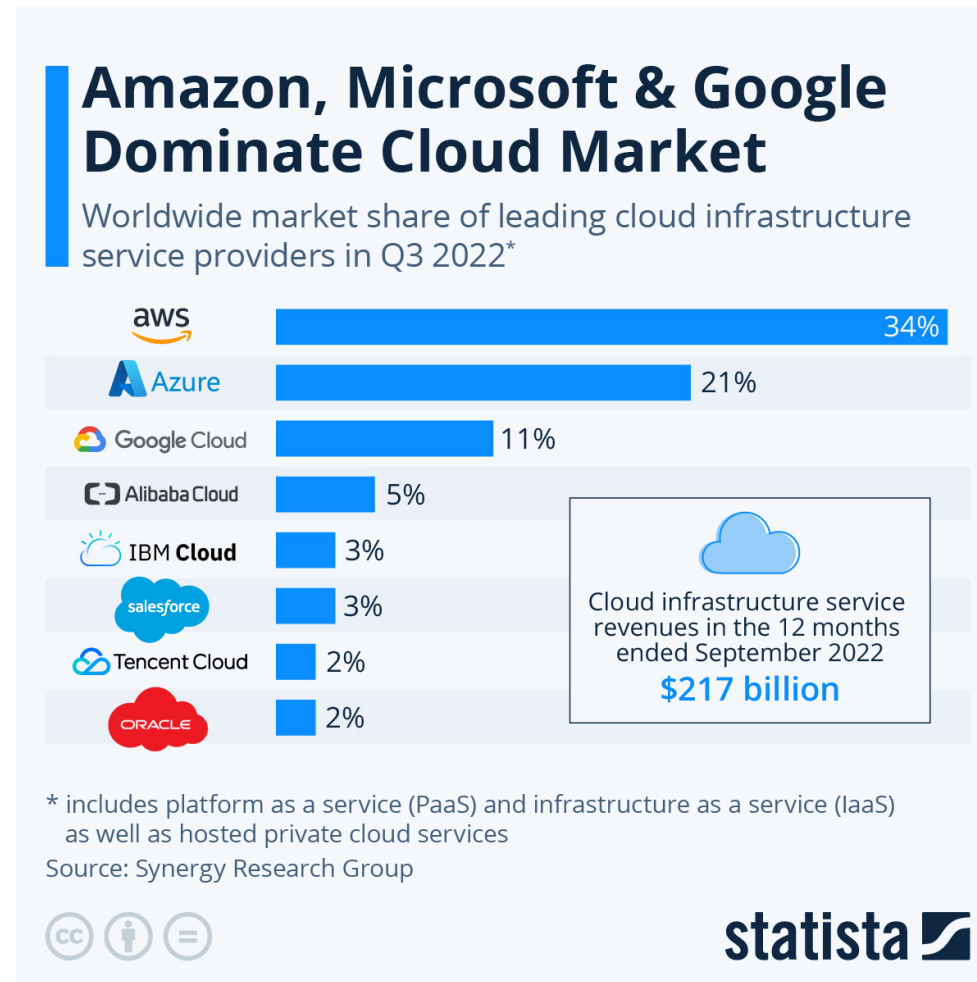
Vietnamese cloud service provider:

- [FPT Smart Cloud](#)



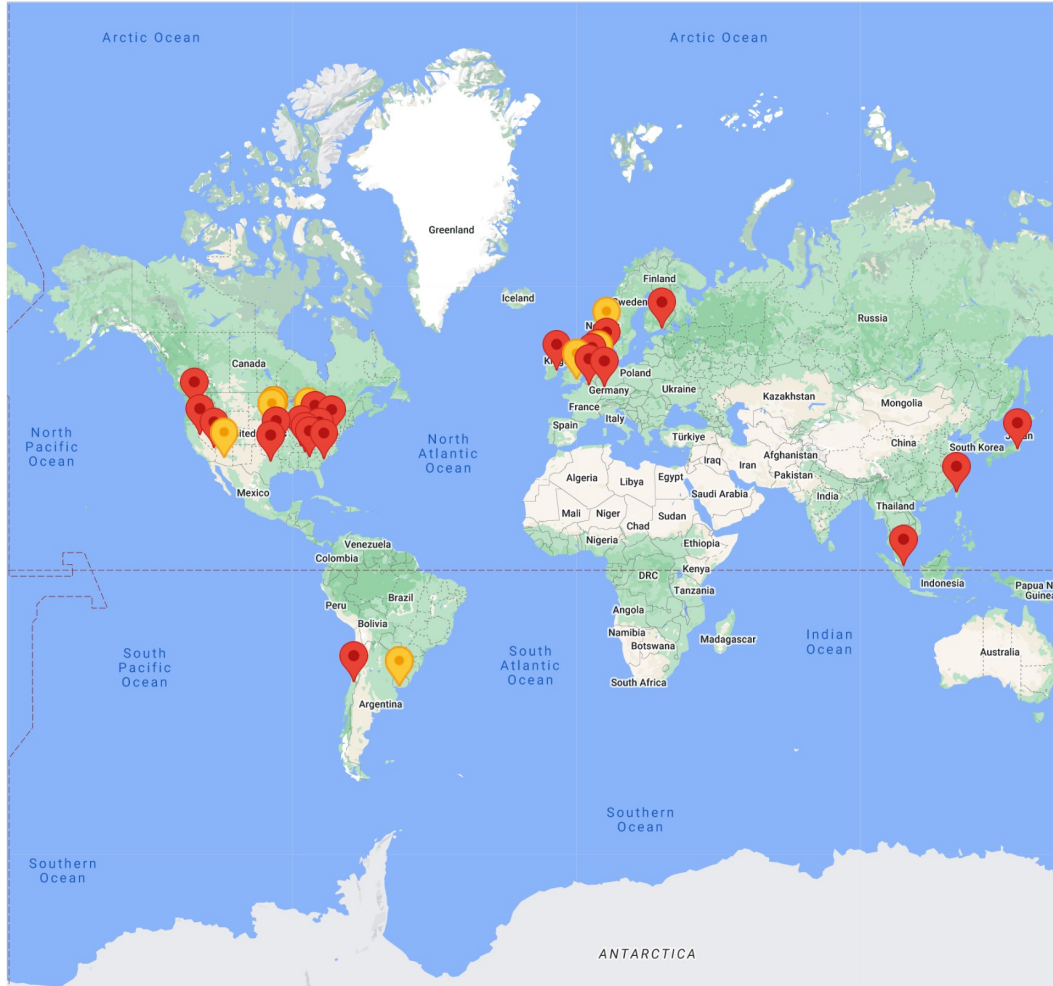
Introduction

❖ Getting Started



Introduction

❖ Getting Started



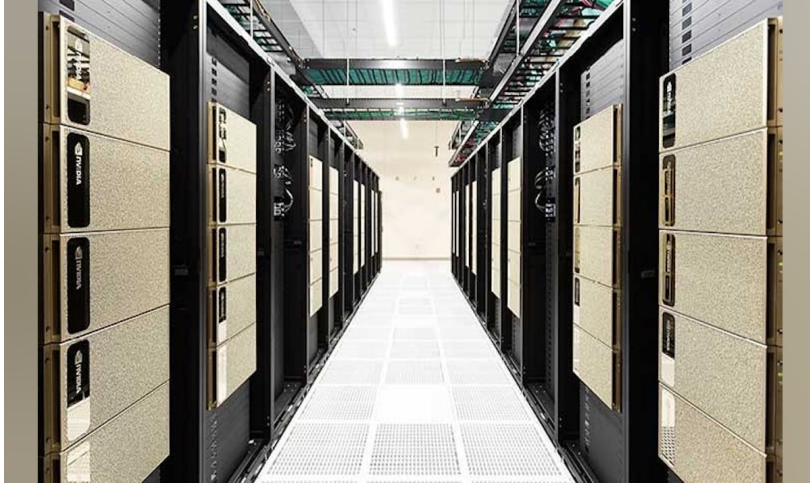
Google Cloud Data Center Map Distribution



AWS Data Center Map Distribution

Introduction

❖ Data Center



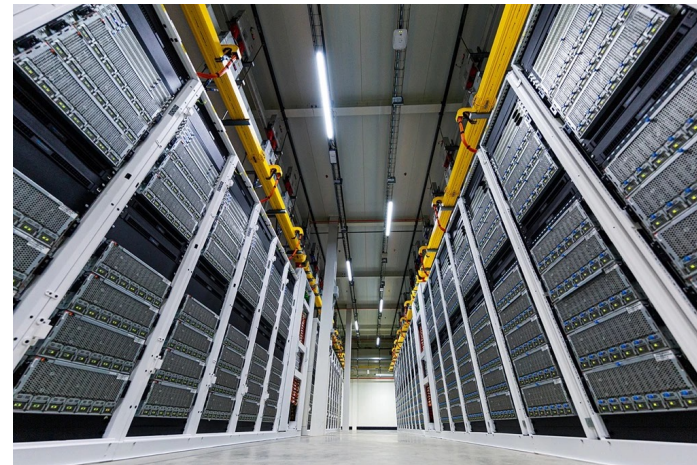
NVIDIA Data Center



AWS Data Center



FPT Data Center



Google Data Center

Introduction

❖ Cloud Services

Cloud Comparison Cheat Sheet blog.bytebytego.com

aws	Azure	Google Cloud	ORACLE CLOUD
Elastic Compute Cloud (EC2)	Virtual Machine	Compute Engine	Virtual Machine
Elastic Kubernetes Service (EKS)	Azure Kubernetes Service (AKS)	Google Kubernetes Engine (GKE)	Oracle Container Engine
Lambda	Azure Functions	Cloud Functions	OCI Functions
Simple Storage Service (S3)	Blob Storage	Cloud Storage	Object Storage
Elastic Block Store	Managed Disk	Persistent Disk	Persistent Volume
Elastic File System	File Storage	File Store	File Storage
Virtual Private Cloud	Virtual Network	Virtual Private Cloud	Virtual Cloud Network
Route 53	DNS	Cloud DNS	DNS
Elastic Load Balancing	Load Balancer	Cloud Load Balancing	Load Balancer
Web Application Firewall	Web Application Firewall	Cloud Armor	Web Application Firewall
RDS	SQL Database	Cloud SQL	ATP
DynamoDB	Cosmos DB	Firebase Realtime Database	NoSQL Database
Redshift	Synapse Analytics	BigQuery	Autonomous Data Warehouse
Elastic MapReduce	HDInsight	Dataproc	Big Data
Kinesis	Streaming Analytics	Dataflow	Streaming
SageMaker	Machine Learning	Vertex AI	Data Science
Glue	Data Factory	Data Fusion	Data Integration
EventBridge	Event Grid	Eventarc	Events
Simple Queuing Service	Storage Queues	Pub/Sub	Streaming
Simple Notification Service	Service Bus	Firebase Cloud Messaging	Notifications
CloudWatch	Monitor	Cloud Monitoring	Monitoring
CloudFormation	Resource Manager	Deployment Manager	Resource Manager
IAM	Active Directory	Cloud Identity	IAM
KMS	Key Vault	Cloud KMS	Vault

Introduction

❖ Why do we need cloud?

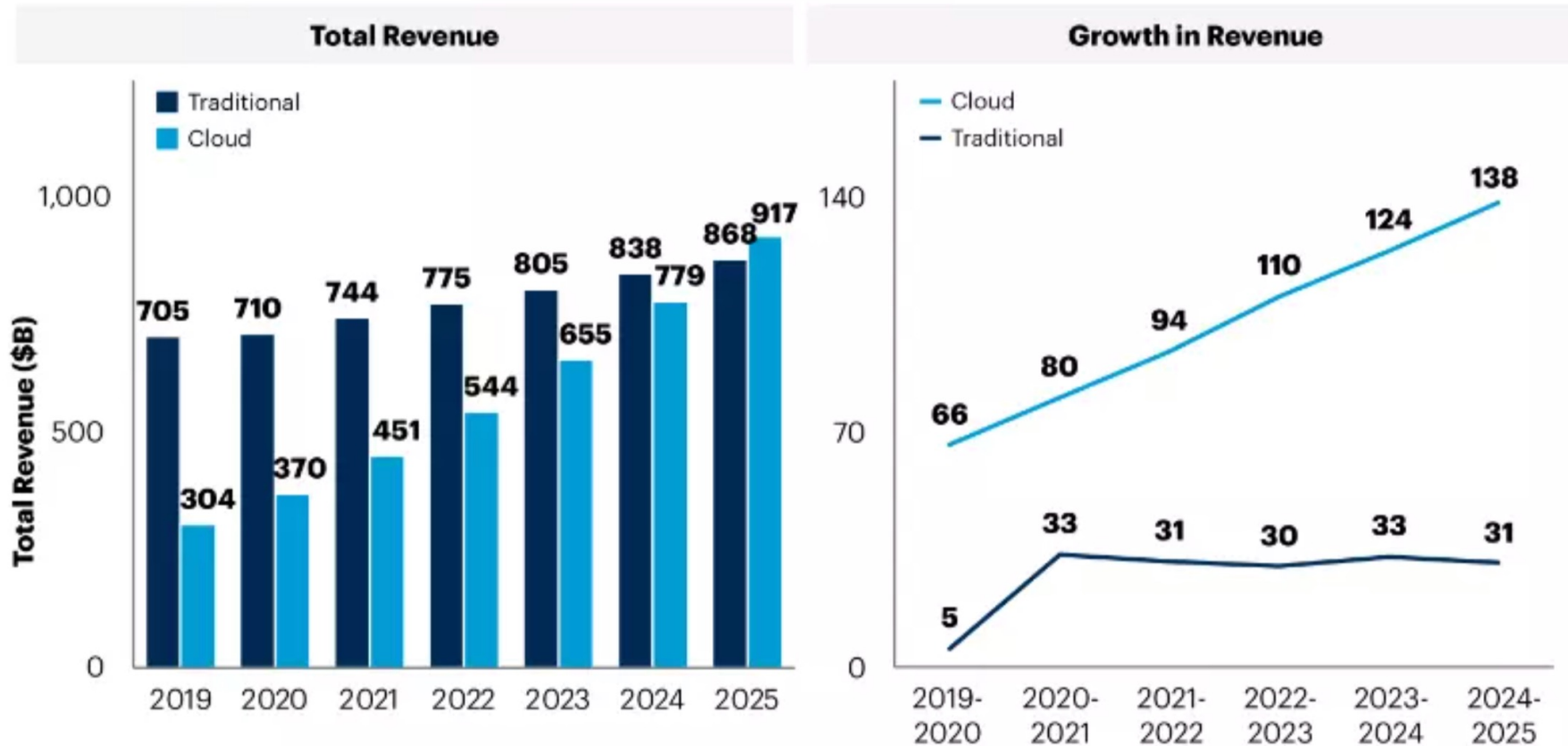
- Economy - affordable cost
- Security - system ready to use
- Scalability - autoscaling features
- Focus - focus on business only
- Agility - easy to create and delete resources
- Maintenance - only cloud provider will handle that



	On-premises	Cloud
Location	On your own site	On Provider's servers
Accessibility	Very high	Very High
Use Cases	For highly confidential organizations	For variable demands and dynamic workloads
Technical Involvement	Very high	Very low
Upfront cost	Extremely high	Very affordable fee structure
Maintenance responsibility	The onus is on the organization	Shared responsibility model
Control & Customization	Organizations have A-Z control over everything	Requires cloud expertise to explore all the customization options.
Security & Compliance	Can be curated on individual needs, but very high responsibility	Robust security and compliance
Resource Deployment	Very time-consuming and involves high downtime	Involves less time and negligible downtime
Vendor lock-ins	No concerns	Low to moderate
Scalability	Too complicated	Easy and cost-effective

Introduction

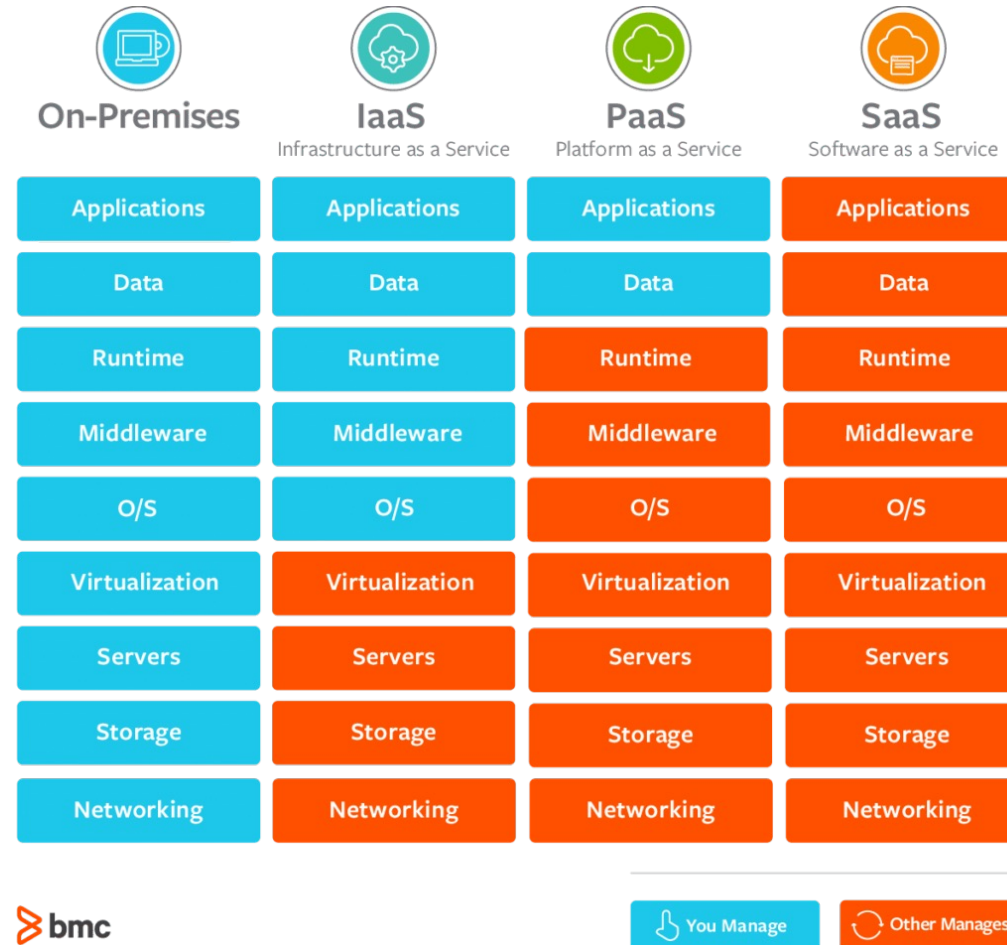
❖ Cloud Revenue



Source: Gartner
758067_C

Introduction

❖ Cloud Service Models



Introduction

❖ Calculate the cost

Why do we need cloud calculator?

- Estimate the cost services in cloud
- Plan ahead when running resources
- Control the billing in organization

Cloud Calculator

- [AWS calculator](#)
- [GCP calculator](#)
- [Azure calculator](#)

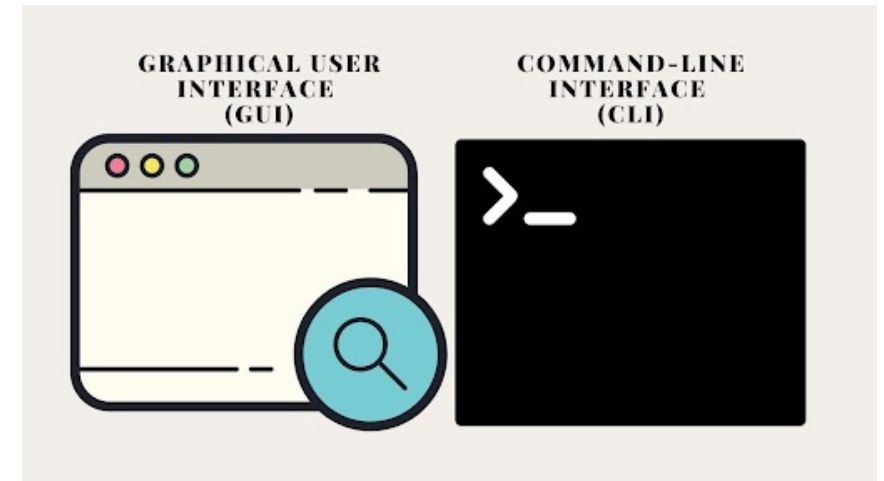


Command Line (CLI) with Cloud

Command line (CLI) with Cloud

❖ What is command line interface (CLI)?

- A command-line interface is a means of interacting with a computer program by inputting lines of text called command-lines
- Tell computer do to something as copy files, move files, or create folders ...
- A shell is a **program** that acts as command-line interpreter. It **processes commands** and **outputs the results**. It interprets and processes the commands entered by the user.
 - BASH (Bourne Again SHell) – It is the most widely used shell in Linux systems.
 - [Z Shell](#)



Command line (CLI) with Cloud

❖ What is command line interface (CLI)?

```
override@Atul-HP: ~  
override@Atul-HP:~$ ls -l  
total 212  
drwxrwxr-x 5 override override 4096 May 19 03:45 acadenv  
drwxrwxr-x 4 override override 4096 May 27 18:20 acadview_demo  
drwxrwxr-x 12 override override 4096 May 3 15:14 anaconda3  
drwxr-xr-x 6 override override 4096 May 31 16:49 Desktop  
drwxr-xr-x 2 override override 4096 Oct 21 2016 Documents  
drwxr-xr-x 7 override override 4096 Jun 1 13:09 Downloads  
-rw-r--r-- 1 override override 8980 Aug 8 2016 examples.desktop  
-rw-rw-r-- 1 override override 45005 May 28 01:40 hs_err_pid1971.log  
-rw-rw-r-- 1 override override 45147 Jun 1 03:24 hs_err_pid2006.log  
drwxr-xr-x 2 override override 4096 Mar 2 18:22 Music  
drwxrwxr-x 21 override override 4096 Dec 25 00:13 Mydata  
drwxrwxr-x 2 override override 4096 Sep 20 2016 newbin  
drwxrwxr-x 5 override override 4096 Dec 20 22:44 nltk_data  
drwxr-xr-x 4 override override 4096 May 31 20:46 Pictures  
drwxr-xr-x 2 override override 4096 Aug 8 2016 Public  
drwxrwxr-x 2 override override 4096 May 31 19:49 scripts  
drwxr-xr-x 2 override override 4096 Aug 8 2016 Templates  
drwxrwxr-x 2 override override 4096 Feb 14 11:22 test  
drwxr-xr-x 2 override override 4096 Mar 11 13:27 Videos  
drwxrwxr-x 2 override override 4096 Sep 1 2016 xdm-helper  
override@Atul-HP:~$
```

Bash Shell

```
wesbos@wb: ~/Dropbox/0219-command-line-power-user (zsh)  
+ ~/Dropbox z command  
+ ~/Dropbox/0219-command-line-power-user master cloudapp err.log  
Uploaded to http://wes.io/aIcd.  
+ ~/Dropbox/0219-command-line-power-user master trash err.log  
+ ~/Dropbox/0219-command-line-power-user master git add -A  
+ ~/Dropbox/0219-command-line-power-user master git commit -m "removed log"  
[master 9313b5e] removed log  
1 file changed, 0 insertions(+), 0 deletions(-)  
delete mode 100644 err.log  
+ ~/Dropbox/0219-command-line-power-user master npm i.....
```

Z Shell

Command line (CLI) with Cloud

❖ Basic command line - Navigation

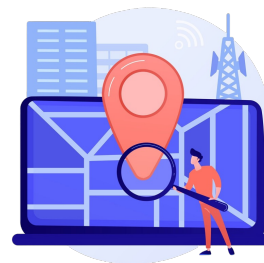
Function	Command	Description
Navigation	> pwd	Show the path of current directory
Navigation	> ls	List all contents in current directory
Navigation	> cd <directory name>	Change to specified directory



Command line (CLI) with Cloud

❖ Basic command line – Manipulate files

Function	Command	Description
Manipulate files	> mkdir <name>	Make new directories according to <name>
Manipulate files	> rmdir <name>	Delete directories according to <name>
Manipulate files	> rm <filename>	Remove file according to <filename>
Manipulate files	> cp <file1> <file2>	Make a copy of <file1> in location with the name of <file2>
Manipulate files	> mv <file1> <file2>	Move <file1> to a new location <file2>



Command line (CLI) with Cloud

❖ Basic command line – Reading files

Function	Command	Description
Reading files	> less <filename>	Show 1 page. Space bar with go to the next page
Reading files	> cat <filename>	Print the whole file in terminal
Reading files	> head <filename>	Display first 10 lines of <filename>
Reading files	> tail <filename>	Display last 10 lines
Reading files	> grep <word>	Search specific word in terminal



Command line (CLI) with Cloud

❖ Basic command line – Reading files

Function	Command	Description
Other	> history	Show all command history
Other	> clear	clear the content display in terminal
Other	> whoami	Show the name of current user
Other	> cal	Show calendar
Other	> shutdown	Shutdown a computer
Other	> reboot	Restart a computer
Other	> htop	Monitor process viewer and process manager

Command line (CLI) with Cloud

❖ Basic command line – Reading files

```
shum@sol:~$ ls -l
total 20
drwx----- 2 shum staff 4096 Jan 16 22:04 Mail
drwx----- 3 shum staff 4096 Jan 16 14:15 csc128
drwxr-xr-x  2 shum staff 4096 Jan 13 16:42 public
drwxr-xr-x  2 shum staff 4096 Jan 16 14:07 public_html
-rw-r--r--  1 shum staff 628 Jan 15 20:04 verse
```

Annotations for the `ls -l` output:

- file type**: Indicated by the first character of the permissions (e.g., `d` for directory, `-` for regular file).
- user (owner) name**: The user name following the permissions (e.g., `shum`).
- group name**: The group name following the user name (e.g., `staff`).
- size**: The file size in bytes (e.g., `4096`).
- date/time last modified**: The date and time the file was last modified (e.g., `Jan 16 22:04`).
- filename**: The name of the file (e.g., `Mail`).
- permissions**: Indicated by the next nine characters (e.g., `drwx-----`).
- number of hard links**: The number following the permissions (e.g., `2`).
- other (everyone) permissions**: Indicated by the last three characters of the permissions (e.g., `---`).
- group permissions**: Indicated by the middle three characters of the permissions (e.g., `---`).
- user permissions**: Indicated by the first three characters of the permissions (e.g., `drwx`).
- executable**: Indicated by the `x` in the permissions (e.g., `x` in `drwx`).
- writable**: Indicated by the `w` in the permissions (e.g., `w` in `drwx`).
- readable**: Indicated by the `r` in the permissions (e.g., `r` in `drwx`).

Owner			Group			Others		
r	w	x	r	w	x	r	w	x
4	2	1	4	2	1	4	2	1

Owner Group Others

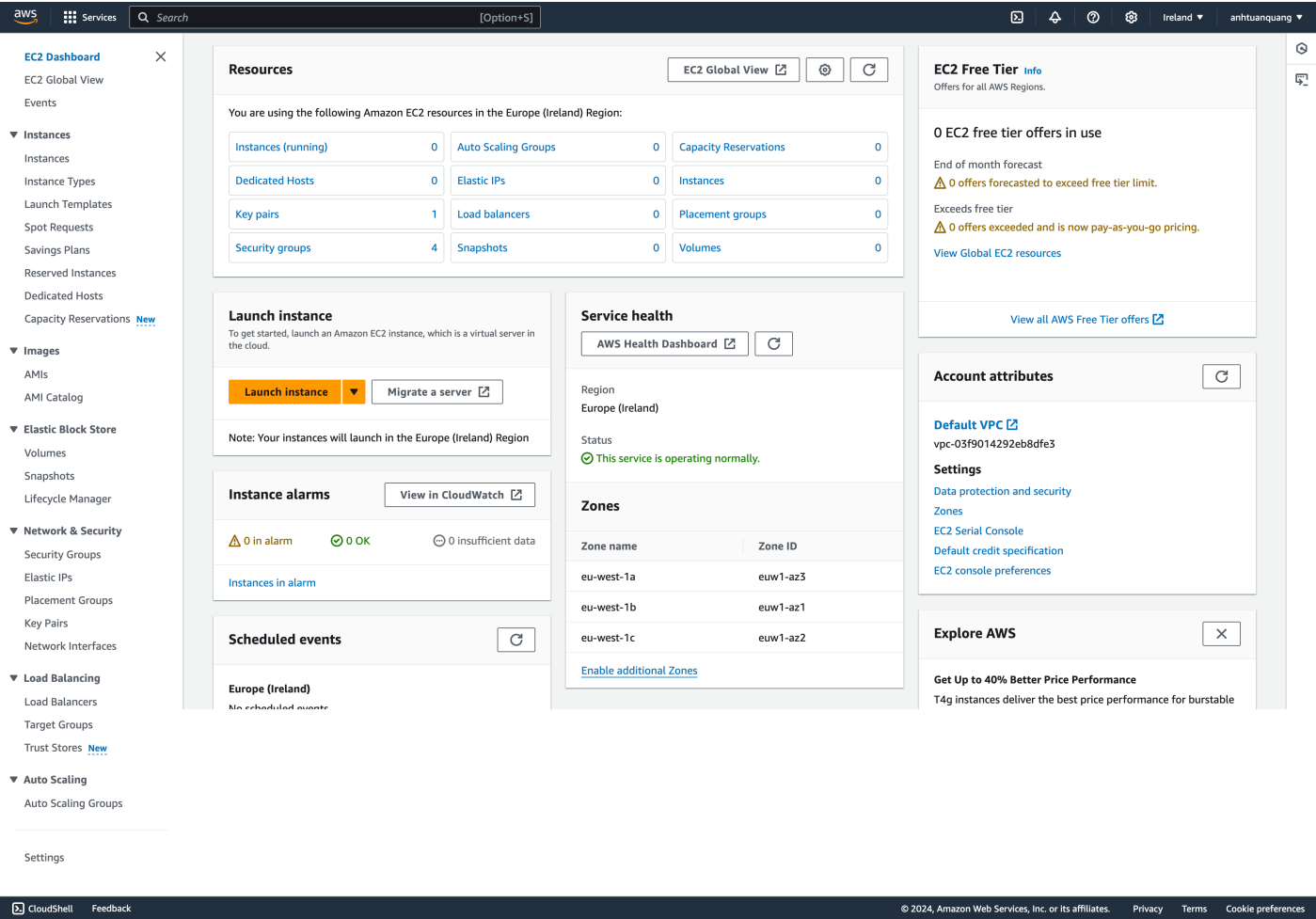
> `chmod 700 <filename>`

Exercise

Cloud Deploy

Cloud Deploy

❖ AWS Console



Cloud Deploy

❖ GCP Console

Google Cloud My First Project Search (/) for resources, docs, products, and more Search

Recommended based on your interest in Web, Mobile, Game, Storage

Pre-built solution templates ⓘ

- Deploy a three-tier web app**
Web app, rich media site, ecommerce website, database-backed website
- Create a dynamic web app with Python and JS**
Dynamic web app, ecommerce website, database backend, django rest framework,...
- Create a serverless ecommerce platform**
Ecommerce web app, database backend, django

[View all solutions](#)

Products

- Create a VM**
Compute Engine
- Create a database**
Cloud SQL
- Run containerized apps**
Cloud Run
- Create a storage bucket**
Cloud Storage
- Create a containerized app**
Kubernetes Engine
- Access services**
APIs
- Create a network**
VPC
- Build a mobile or web app**
Firebase

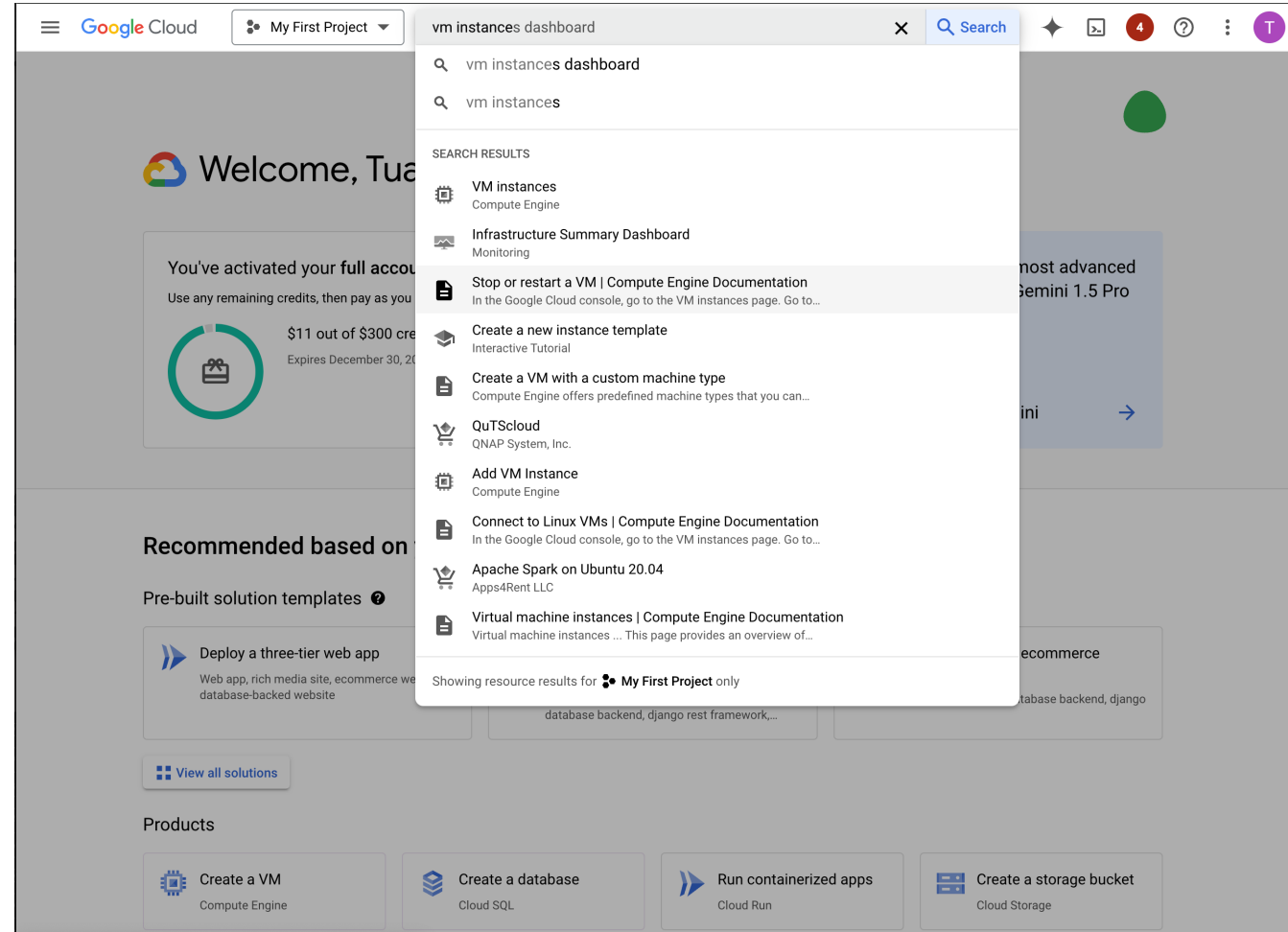
[View all products](#) [View all APIs](#) [Compare with AWS and Azure](#)

Interactive tutorials

- Building a custom website with Google...**
15 min
- Deploy a containerized web application with GKE**
10 min
- Creating and deploying a Cloud Run service**
10 min
- Create a MySQL instance by using Cloud SQL**
15 min
- Create a Cloud Storage bucket**
5 min
- Deploy a WordPress website by using Googl...**
15 min
- Create a Cloud Storage bucket using a client...**
10 min

Cloud Deploy

❖ GCP – VM Machine



Cloud Deploy

❖ GCP – VM Machine

Google Cloud | My First Project | vm instance | Search | 4 | ? | T

Compute Engine

VM Manager

- Patch
- OS policies

Bare Metal Solution

- Servers
- Networks
- VRFs **NEW**
- Volumes
- NFS Shares
- Procurements **NEW**
- Maintenance Events **NEW**

Settings

VM instances | **CREATE INSTANCE** | IMPORT VM | REFRESH | LEARN

INSTANCES | OBSERVABILITY | INSTANCE SCHEDULES

VM instances

Filter Enter property name or value

Status	Name ↑	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☒	tuan-machine	us-central1-c			10.128.0.2 (nic0)	34.56.61.134 (nic0)	SSH ▾

Related actions | HIDE

- Explore Backup and DR** **NEW**
Back up your VMs and set up disaster recovery
- View billing report**
View and manage your Compute Engine billing
- Monitor VMs**
View outlier VMs across metrics like CPU and network
- Explore VM logs**
View, search, analyze, and download VM instance logs
- Set up firewall rules**
Control traffic to and from a VM instance
- Patch management**
Schedule patch updates and view patch compliance on VM instances

❖ GCP – VM Machine (Machine Configuration)

- VM basics
instance-20241022-053051,
us-central1
- Machine configuration
e2-medium
- OS and storage
Debian GNU/Linux 12
(bookworm)
- Networking
1 network interface
- Observability
- Security
- Advanced

Machine configuration

NEW: Google Axion virtual machines in Preview

Try Google Axion Arm-based Processors. Sign up for early access now

SIGN UP

General purpose

Compute optimizedMemory optimizedStorage optimizedGPUs

Machine types for common workloads, optimized for cost and flexibility

	Series	Description	vCPUs	Memory	Platform
☐	C4	Consistently high performance	2 - 192	4 - 1,488 GB	Intel Emerald Rapids
☐	N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald Rapids
☐	C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire Rapids
☐	C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒	E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Based on availability
☐	N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade and Ice Lake
☐	N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD EPYC
☐	T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra Arm
☐	T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD EPYC Milan
☐	N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Skylake

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

PRESET

CUSTOM

e2-medium (2 vCPU, 1 core, 4 GB memory)

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly esti
2 vCPU + 4 GB memory	\$2
10 GB balanced persistent disk	\$
Total	\$2

[Compute Engine pricing](#)

LESS

Cloud Deploy

❖ GCP – VM Machine (OS and storage)

← Create an instance

✦ CREATE VM FROM...

EQUIVALENT CODE <I

• VM basics
instance-20241022-053051,
us-central1

• Machine configuration
e2-medium

• OS and storage
Debian GNU/Linux 12
(bookworm)

• Networking
1 network interface

• Observability

• Security

• Advanced

Operating system and storage

Name

instance-20241022-053051

Type

New balanced persistent disk

Size

10 GB

Snapshot schedule ?

No schedule selected

License type ?

Free

Image

Debian GNU/Linux 12 (bookworm)

CHANGE

Additional storage and VM backups

+ ADD NEW DISK

+ ATTACH EXISTING DISK

+ ADD LOCAL SSD

Backup plan PREVIEW

Secure your backups against deletion through backup vault storage and enable centralized backup management across projects. Managed by Backup and DR Service, a separate service from Compute Engine with independent certifications and accreditation. [Learn more](#)

Backup plan

SELECT A PLAN ?

Container ?

Deploy a container image to this VM instance

DEPLOY CONTAINER

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Total	\$25.46

[Compute Engine pricing](#)

^ LESS

CREATE

CANCEL

EQUIVALENT CODE

Cloud Deploy

❖ GCP – VM Machine (Network)

Google Cloud

My First Project

Search (/) for resources, docs, products, and more

Search

4

?

T

Create an instance

CREATE VM FROM...

EQUIVALENT CODE

VM basics
instance-20241022-060545,
us-central1

Machine configuration
e2-medium

OS and storage
Debian GNU/Linux 12
(bookworm)

Networking
1 network interface

Observability

Security

Advanced

Firewall

Add tags and firewall rules to allow specific network traffic from the Internet

☐ Allow HTTP traffic

☐ Allow HTTPS traffic

☐ Allow Load Balancer Health Checks

Network tags

Hostname

Set a custom hostname for this instance or leave it default. Choice is permanent

IP forwarding

☐ Enable

Network performance configuration

Network interface card

Network bandwidth

☐ Enable per VM Tier_1 networking performance

Maximum outbound network bandwidth: 2Gbps

VM to Public IP: 2Gbps

Network interfaces

Network interface is permanent

▼ default default IPv4 (10.128.0.0/20)

ADD A NETWORK INTERFACE

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Total	\$25.46

[Compute Engine pricing](#)

LESS

CREATE

CANCEL

EQUIVALENT CODE

❖ GCP – VM Machine

Google Cloud

My First Project

Search (/) for resources, docs, products, and more

Search

4

?

T

Create an instance

CREATE VM FROM...

EQUIVALENT CODE

VM basics

instance-20241022-060907, us-central1

Machine configuration

e2-medium

OS and storage

Debian GNU/Linux 12 (bookworm)

Networking

1 network interface

Observability

Security

Advanced

Identify your VM

Name *

instance-20241022-060907

Region *

us-central1 (Iowa)

Region is permanent

Zone *

Any

Google will choose a zone on your behalf, maximizing VM obtainability. Zone is permanent.

MANAGE TAGS AND LABELS

Availability policies

VM provisioning model

Standard

Choose "Spot" to get a discounted, preemptible VM. Otherwise, stick to "Standard". [Learn more](#)

VM PROVISIONING MODEL ADVANCED SETTINGS

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Total	\$25.46

[Compute Engine pricing](#)

LESS

Create instance - Final Step

CREATE

CANCEL

EQUIVALENT CODE

❖ GCP – VM Machine

Google Cloud

My First Project

Search (/) for resources, docs, products, and more

Search

4

?

T

Create an instance

CREATE VM FROM...

EQUIVALENT CODE

VM basics

instance-20241022-060907,
us-central1

Machine configuration

e2-medium

OS and storage

Debian GNU/Linux 12
(bookworm)

Networking

1 network interface

Observability

Security

Advanced

Identify your VM

Name *

instance-20241022-060907

Region *

us-central1 (Iowa)

Region is permanent

Zone *

Any

Google will choose a zone on your behalf,
maximizing VM obtainability. Zone is permanent.

MANAGE TAGS AND LABELS

Availability policies

VM provisioning model

Standard

Choose "Spot" to get a discounted, preemptible VM. Otherwise, stick to "Standard". [Learn more](#)

VM PROVISIONING MODEL ADVANCED SETTINGS

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Total	\$25.46

[Compute Engine pricing](#)

LESS

Create instance - Final Step

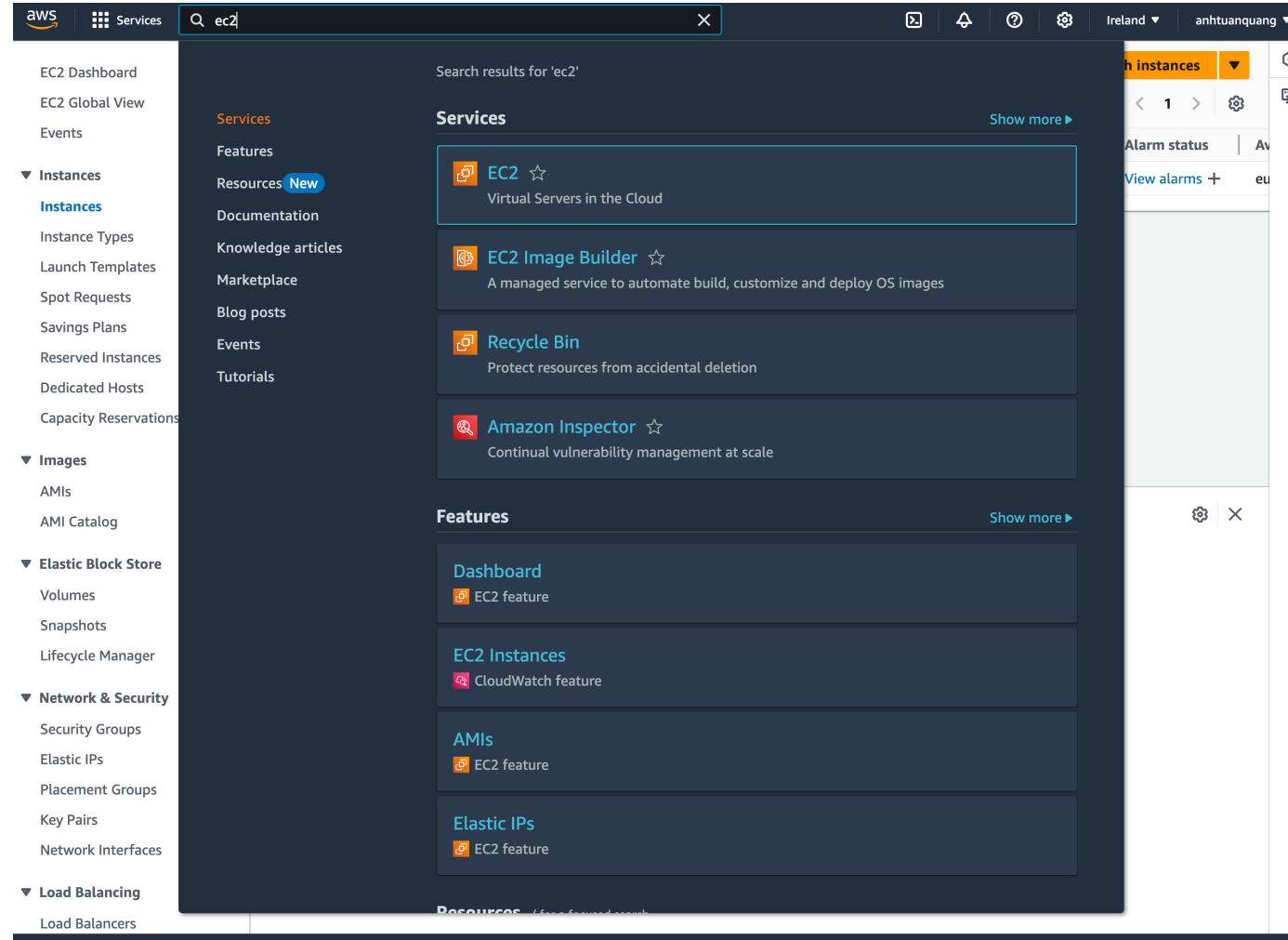
CREATE

CANCEL

EQUIVALENT CODE

Cloud Deploy

❖ AWS – EC2 (create virtual server)



Cloud Deploy

❖ AWS – EC2 (create virtual server)

The screenshot displays the AWS Management Console's EC2 Dashboard for the Europe (Ireland) Region. The left-hand navigation pane includes sections for EC2 Dashboard, Instances, Images, Elastic Block Store, and Network & Security. The main content area is divided into several panels: a 'Resources' panel showing counts for various EC2 resources; a 'Launch instance' panel with a prominent orange button and a green arrow pointing to it; a 'Service health' panel indicating normal operation; an 'Instance alarms' panel showing no active alarms; and a right-hand sidebar with 'EC2 Free Tier' information and 'Account attributes'.

Resources

You are using the following Amazon EC2 resources in the Europe (Ireland) Region:

Instances (running)	1	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	1
Key pairs	1	Load balancers	0	Placement groups	0
Security groups	5	Snapshots	0	Volumes	1

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

[Launch instance](#) [Migrate a server](#)

Note: Your instances will launch in the Europe (Ireland) Region

Service health

Region: Europe (Ireland)

Status: ✔ This service is operating normally.

Zones

Zone name	Zone ID
eu-west-1a	euw1-az3
eu-west-1b	euw1-az1

Instance alarms

⚠ 0 in alarm ✔ 0 OK ⌵ 0 insufficient data

[Instances in alarm](#)

EC2 Free Tier

Offers for all AWS Regions.

0 EC2 free tier offers in use

End of month forecast
⚠ 0 offers forecasted to exceed free tier limit.

Exceeds free tier
⚠ 0 offers exceeded and is now pay-as-you-go pricing.

[View Global EC2 resources](#)

[View all AWS Free Tier offers](#)

Account attributes

[Default VPC](#)
vpc-03f9014292eb8dfe3

Settings

- [Data protection and security](#)
- [Zones](#)
- [EC2 Serial Console](#)
- [Default credit specification](#)
- [EC2 console preferences](#)

Cloud Deploy

❖ AWS – EC2 (configure virtual server and launch)

[EC2](#) > ... > Launch an instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Li

SUS

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI

ami-00385a401487aefa4 (64-bit (x86), uefi-preferred) / ami-08c208c3af7f04012 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.6.2...[read more](#)

ami-00385a401487aefa4

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch instance

Preview code

Cloud Deploy

❖ Connect virtual machine with SSH terminal

What is ssh-keygen?

- SSH-keygen is a tool used to create authentication key pairs for SSH (Secure Shell)



```
→ Desktop ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/tuanquang/.ssh/id_ed25519):
```

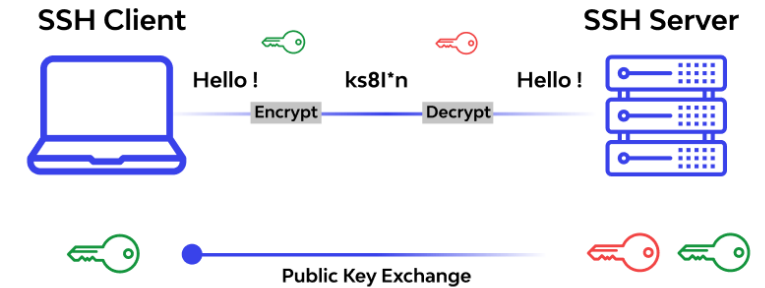
ssh-keygen in Linux

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\tuanquang> ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\tuanquang\.ssh\id_ed25519):
```

ssh-keygen in window



Cloud Deploy

❖ VS Code - SSH Remote

The screenshot displays the Visual Studio Code interface with the Extensions Marketplace open. The search bar is set to 'ssh'. The 'Remote - SSH' extension by Microsoft is selected, showing its details. A tooltip for 'SSH FS v1.26.1' is visible over the 'Remote - SSH' extension card. The main panel for 'Remote - SSH' shows its description, features, and a list of categories. The bottom panel shows a code editor with a Python script.

EXTENSIONS: MARKETPLACE

ssh

Remote - SSH 92ms
Open any folder on a remote machine using SSH and take a...
Microsoft

SSH FS 597K 4.5
File system, terminal and task provider using SSH
Kelvin Schoofs

Remote - SSH: Editing Configuration Files
Edit SSH configuration files
Microsoft

Remote X11 (SSH) 165K 5
Enables X11 forwarding with the Remote - SSH extension.
Joel Spadin

Save as Root in Remote - SSH 193K 5
Edit configuration files with root privileges while using the R...
yy0931

SSH Client 91K 4.5
SSH Client and Database manager for MySQL/MariaDB, Pos...
Weijian Chen

SSH Terminal 45K 5
SSH Terminal
cooper

SSH Tools 17K 5
XPLOT Tools for vscode;SSH,FTP,Remote
XPLOT

Terminal SSH 37K 1

Remote - SSH v0.115.0
Microsoft | 23,297,744 | ★★★★★ (187)
Open any folder on a remote machine using SSH and take advantage of VS Code's full feature set.

SSH FS v1.26.1
File system, terminal and task provider using SSH
Hold Option key to mouse over

Visual Studio Code Remote - SSH

The **Remote - SSH** extension lets you use any remote machine with a SSH server as your development environment. This can greatly simplify development and troubleshooting in a wide variety of situations. You can:

- Develop on the same operating system you deploy to or use larger, faster, or more specialized hardware than your local machine.
- Quickly swap between different, remote development environments and safely make updates without worrying about impacting your local machine.
- Access an existing development environment from multiple machines or locations.
- Debug an application running somewhere else such as a customer site or in the cloud.

No source code needs to be on your local machine to gain these benefits since the extension runs commands and other extensions directly on the remote machine. You can open any folder on the remote machine and work with it just as you would if the folder were on your own machine.

Categories
Other

Resources
[Marketplace](#)
[Issues](#)
[Repository](#)
[License](#)
[Microsoft](#)

More Info

Published	2019-05-02, 11:40:34
Last released	2024-10-07, 08:21:21
Last updated	2024-10-04, 17:11:32
Identifier	ms-vscode-

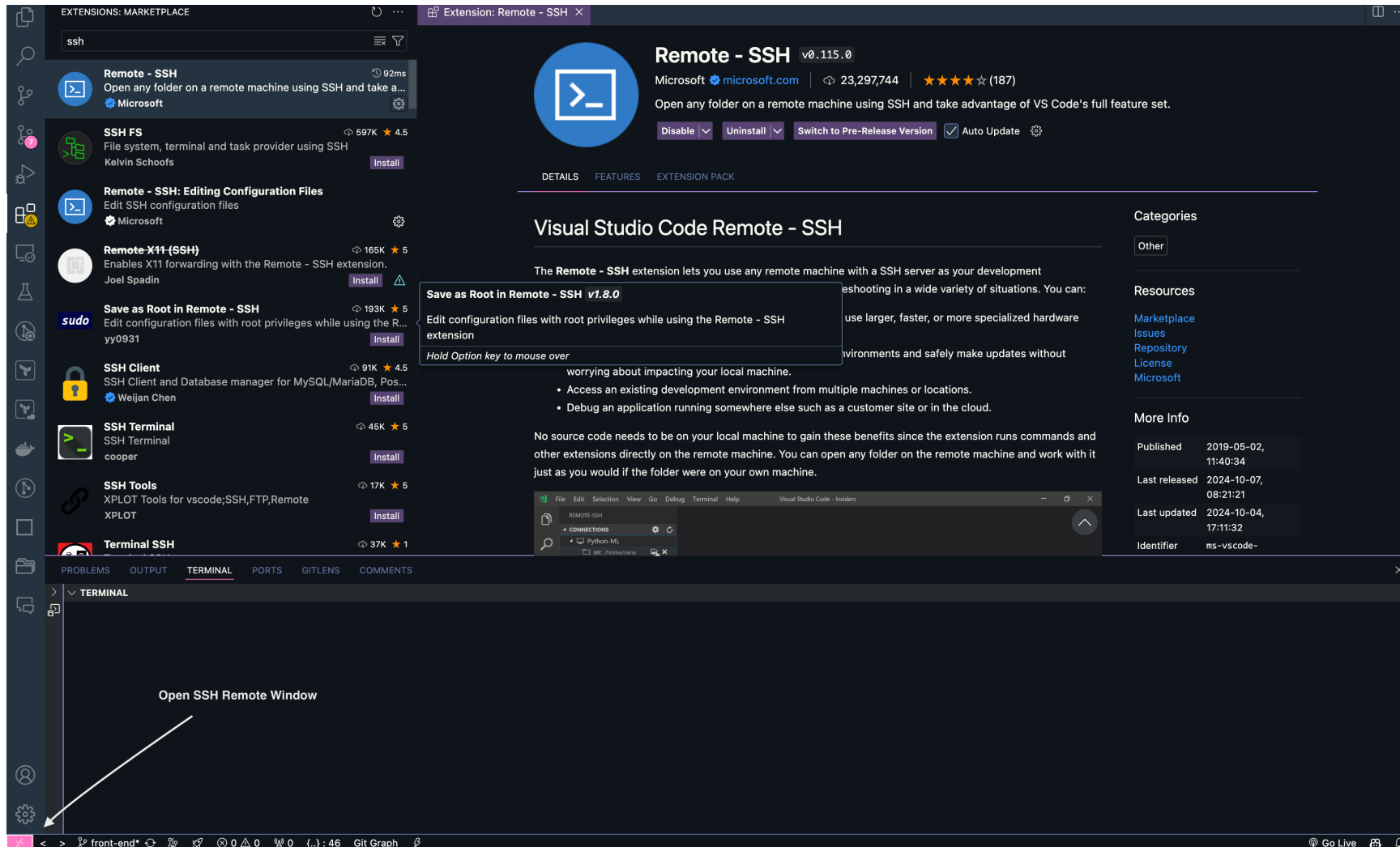
File Edit Selection View Go Debug Terminal Help
main.py - src [SSH: Python-ML] - Visual Studio Code - Insiders

EXPLORER
main.py x
main.py > ...
main.py > ...
main.py > ...

```
1 import time  
2 from sklearn import svm
```

Cloud Deploy

❖ VS Code - SSH Remote



Cloud Deploy

❖ VS Code - SSH Remote

The screenshot displays the Visual Studio Code interface with the 'Remote - SSH' extension page open in the Extensions Marketplace. The left sidebar shows a list of extensions related to SSH, including 'Remote - SSH', 'SSH FS', 'Remote - SSH: Editing Configuration Files', 'Remote - X11 (SSH)', 'Save as Root in Remote - SSH', 'SSH Client', 'SSH Terminal', 'SSH Tools', and 'Terminal SSH'. The main panel shows the details for the 'Remote - SSH' extension by Microsoft, version 0.115.0, with 23,297,744 downloads and a 4.5-star rating. The extension description states: 'Open any folder on a remote machine using SSH and take advantage of VS Code's full feature set.' Below the description, there are tabs for 'DETAILS', 'FEATURES', and 'EXTENSION PACK'. The 'FEATURES' tab is active, showing a list of features: 'The Remote - SSH extension lets you use any remote machine with a SSH server as your development environment in a wide variety of situations. You can: use larger, faster, or more specialized hardware environments and safely make updates without worrying about impacting your local machine.' A list of features includes: 'Access an existing development environment from multiple machines or locations.' and 'Debug an application running somewhere else such as a customer site or in the cloud.' Below the features, it states: 'No source code needs to be on your local machine to gain these benefits since the extension runs commands and other extensions directly on the remote machine. You can open any folder on the remote machine and work with it just as you would if the folder were on your own machine.' A small inset window shows a terminal window with the command 'ssh' entered. The bottom status bar shows the current file is 'front-end*' and the terminal is open.

EXTENSIONS: MARKETPLACE

ssh

Remote - SSH 92ms
Open any folder on a remote machine using SSH and take a...
Microsoft

SSH FS 597K ★ 4.5
File system, terminal and task provider using SSH
Kelvin Schoofs **Install**

Remote - SSH: Editing Configuration Files
Edit SSH configuration files
Microsoft

Remote - X11 (SSH) 165K ★ 5
Enables X11 forwarding with the Remote - SSH extension.
Joel Spadin **Install**

Save as Root in Remote - SSH 193K ★ 5
Edit configuration files with root privileges while using the R...
yy0931 **Install**

SSH Client 91K ★ 4.5
SSH Client and Database manager for MySQL/MariaDB, Pos...
Weijian Chen **Install**

SSH Terminal 45K ★ 5
SSH Terminal
cooper **Install**

SSH Tools 17K ★ 5
XPLOR Tools for vscode,SSH,FTP,Remote
XPLOR **Install**

Terminal SSH 37K ★ 1

Remote - SSH v0.115.0
Microsoft microsoft.com | 23,297,744 | ★★★★★ (187)
Open any folder on a remote machine using SSH and take advantage of VS Code's full feature set.
Disable Uninstall Switch to Pre-Release Version Auto Update

DETAILS FEATURES EXTENSION PACK

Visual Studio Code Remote - SSH

The Remote - SSH extension lets you use any remote machine with a SSH server as your development environment in a wide variety of situations. You can:

- use larger, faster, or more specialized hardware
- environments and safely make updates without worrying about impacting your local machine.

- Access an existing development environment from multiple machines or locations.
- Debug an application running somewhere else such as a customer site or in the cloud.

No source code needs to be on your local machine to gain these benefits since the extension runs commands and other extensions directly on the remote machine. You can open any folder on the remote machine and work with it just as you would if the folder were on your own machine.

Save as Root in Remote - SSH v1.8.0
Edit configuration files with root privileges while using the Remote - SSH extension
Hold Option key to mouse over

Categories
Other

Resources
[Marketplace](#)
[Issues](#)
[Repository](#)
[License](#)
[Microsoft](#)

More Info
Published 2019-05-02, 11:40:34
Last released 2024-10-07, 08:21:21
Last updated 2024-10-04, 17:11:32
Identifier ms-vscode-

PROBLEMS OUTPUT TERMINAL PORTS GITLENS COMMENTS

TERMINAL

Open SSH Remote Window

front-end* 0 0 0 0 {..}: 46 Git Graph

Cloud Deploy

❖ VS Code - SSH Remote

The screenshot displays the Visual Studio Code interface with the 'EXTENSIONS: MARKETPLACE' sidebar on the left. The search bar contains 'ssh', and the 'Remote - SSH' extension by Microsoft is highlighted. Below it, other SSH-related extensions like 'SSH FS', 'Remote - SSH: Editing Configuration Files', 'Remote X11 (SSH)', 'Save as Root in Remote - SSH', 'SSH Client', 'SSH Terminal', 'SSH Tools', and 'Terminal SSH' are listed with their respective install buttons and ratings.

In the center, a dropdown menu titled 'Select an option to open a Remote Window' is open, showing options such as 'Connect to Host...', 'Connect Current Window to Host...', 'Open Container Configuration File', 'New Dev Container...', 'Attach to Running Container...', 'Add Dev Container Configuration Files...', 'Reopen in Container', 'Continue Working in vscode.dev', 'Tunnel', 'GitHub Codespace', and 'Remote Repository'.

On the right, the 'Visual Studio Code Remote - SSH' documentation page is visible, featuring a description of the extension's capabilities and a list of resources.

At the bottom, a terminal window is open, showing the command prompt and the file path 'src /home/sana'.

Visual Studio Code Remote - SSH

The **Remote - SSH** extension lets you use any remote machine with a SSH server as your development environment. This can greatly simplify development and troubleshooting in a wide variety of situations. You can:

- Develop on the same operating system you deploy to or use larger, faster, or more specialized hardware than your local machine.
- Quickly swap between different, remote development environments and safely make updates without worrying about impacting your local machine.
- Access an existing development environment from multiple machines or locations.
- Debug an application running somewhere else such as a customer site or in the cloud.

No source code needs to be on your local machine to gain these benefits since the extension runs commands and other extensions directly on the remote machine. You can open any folder on the remote machine and work with it just as you would if the folder were on your own machine.

Categories

Other

Resources

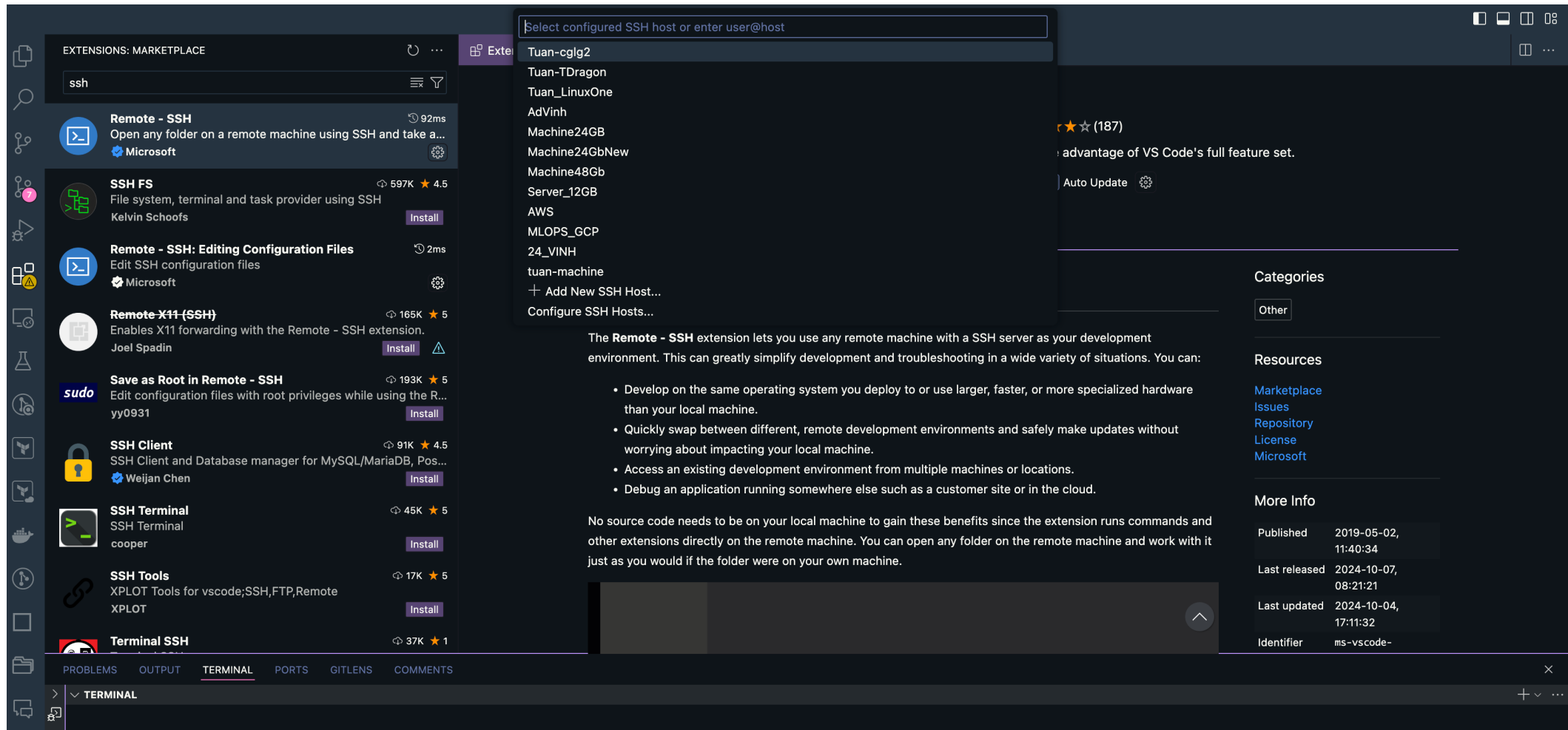
- [Marketplace](#)
- [Issues](#)
- [Repository](#)
- [License](#)
- [Microsoft](#)

More Info

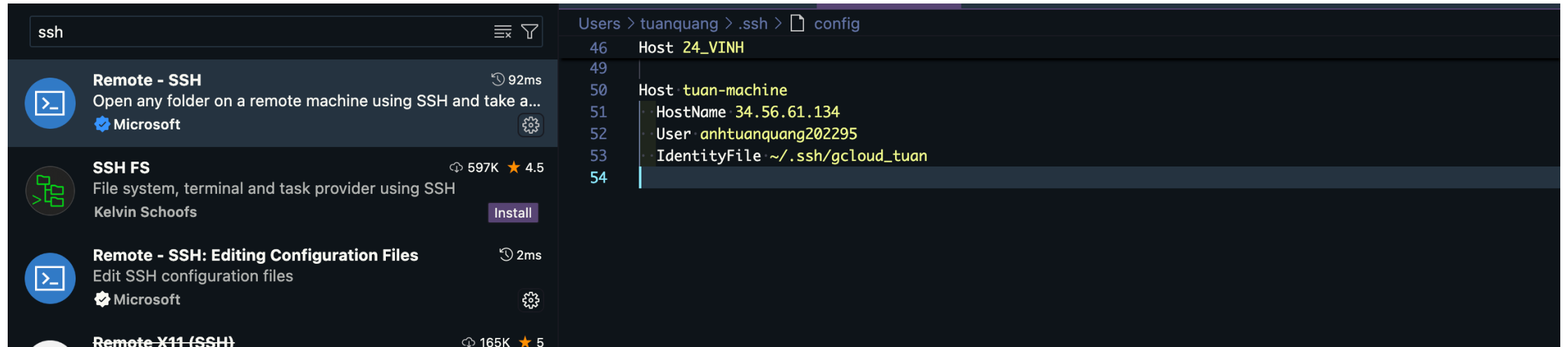
Published	2019-05-02, 11:40:34
Last released	2024-10-07, 08:21:21
Last updated	2024-10-04, 17:11:32
Identifier	ms-vscode-

Cloud Deploy

❖ VS Code - SSH Remote



❖ VS Code - SSH Remote



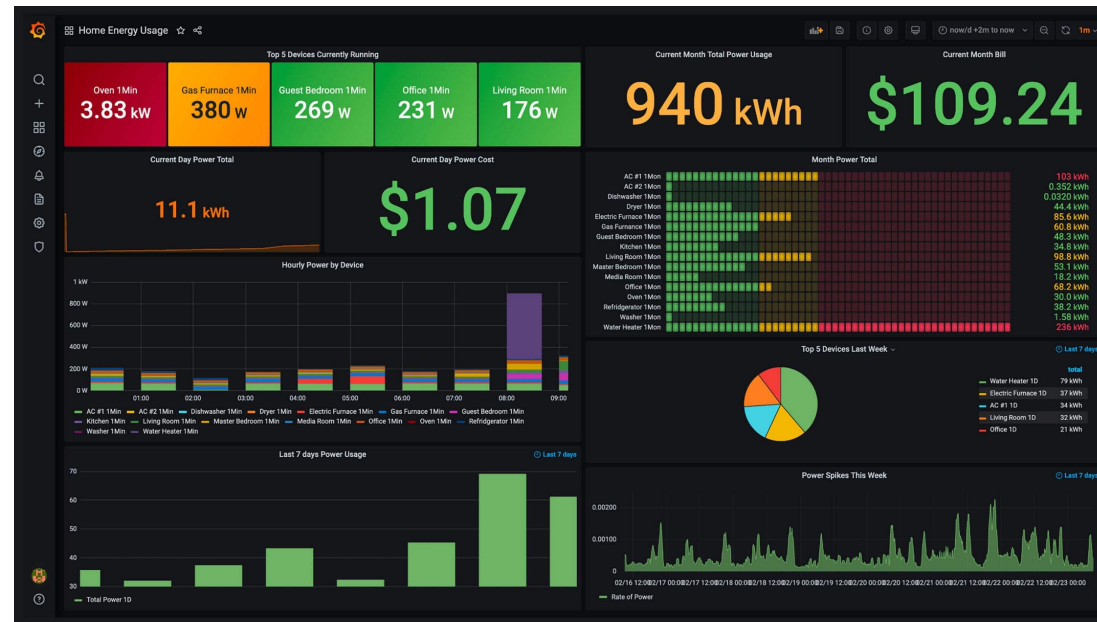
Cloud Monitoring

Cloud Monitoring

❖ Getting Started

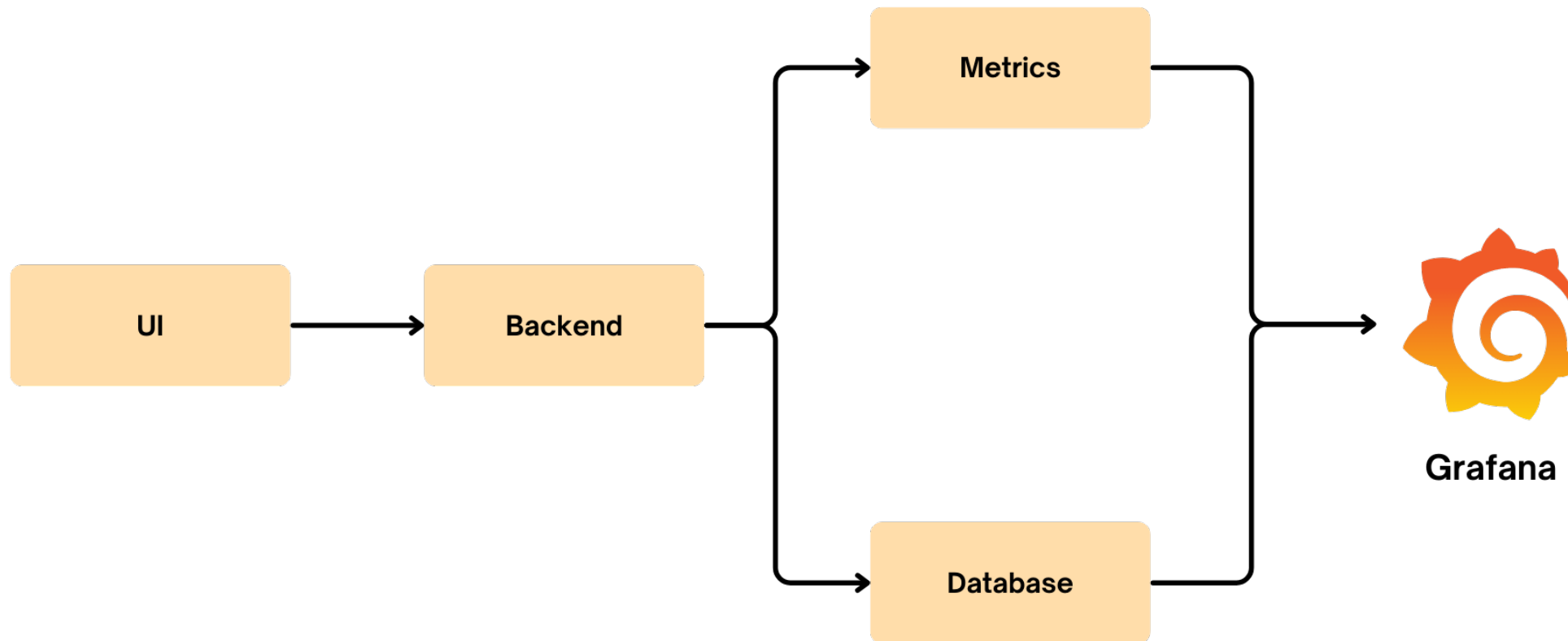
What is Grafana?

- Grafana is a popular open-source analytics and visualization platform. It's primarily used for monitoring and observability purposes in IT and software development environments
- Allow to display all metrics that we want or information from data sources
- Alert when hitting the threshold limitf



Cloud Monitoring

❖ Simple flowchart of application with Grafana



Exercise

Summary

Topic 6 cloud service providers:

- [Amazon Web Services \(AWS\)](#)
- [Microsoft Azure](#)
- [Google Cloud \(GCP\)](#)
- [Alibaba Cloud](#)
- [IBM Cloud](#)
- [Oracle Cloud](#)

Summary



Bash Cheat Sheet



Navigating the File System

cd [directory]	Change directory
pwd	Print working directory
ls [options] [directory]	List directory contents
mkdir [directory]	Create a new directory
rmdir [directory]	Remove a directory
cp [source] [destination]	Copy files or directories
mv [source] [destination]	Move or rename files or directories
rm [options] [file]	Remove files or directories
touch [file]	Create an empty file

Archiving and Compression

tar [options] [files/directories]	Create or extract tar archives
gzip [file]	Compress a file
gunzip [file.gz]	Decompress a gzipped file
zip [archive.zip] [files/directories]	Create a zip archive
unzip [archive.zip]	Extract files from a zip archive



Get more cheat sheets and
other Linux content at
[LinuxStans.com](https://linuxstans.com)

File Manipulation

cat [file]	Output the contents of a file
head [options] [file]	Output the first lines of a file
tail [options] [file]	Output the last lines of a file
less [file]	View the contents of a file interactively
grep [pattern] [file]	Search for a pattern in a file
wc [options] [file]	Count the number of lines, words, or characters in a file

Permissions

chmod [permissions] [file]	Change the permissions of a file or directory
chown [user:group] [file]	Change the owner and group of a file or directory
chgrp [group] [file]	Change the group of a file or directory
umask [mask]	Set the default file permissions for newly created files

Process Management

ps [options]	Display information about active processes
kill [process_ID]	Terminate a process
top	Display and manage the top processes
bg [job_ID]	Move a job to the background
fg [job_ID]	Bring a background job to the foreground

Question

